Homework — 1.2.2 Applications Generation

Name: ____ Date: _ ***Mark:*** ___ / 20

OCR H446 — set after the 1.2.2 lesson. Show your working.

Part A — This week's topic (~14 marks)

1. Classify each of the following as **application software** or **utility software**, and give a one-line reason for each. **[3]** *(AO2)*

> (a) A file compression tool (b) A presentation package (c) A firewall

2. A school is choosing between **open source** and **closed source (proprietary)** office software. State **two** advantages of choosing open source and **one** advantage of choosing closed source. **[3]** *(AO2)*

3. During the **lexical analysis** stage of compilation, what is produced and what is removed? Why is the **symbol table** created at this stage useful to later stages? **[3]** *(AO1)*

4. Explain what happens during **code optimisation** and state why this stage is included even though the program would already run without it. **[2]** *(AO1)*

5. A large program is built from several compiled modules plus a graphics library. Explain the role of the **loader** when the finished executable is run. **[2]** *(AO1)*

6. A games studio updates a shared graphics library after release. Explain why **dynamic linking** lets users benefit from the update without the studio re-releasing the whole program. **[1]** *(AO2)*

Part B — Retrieval: keep it fresh (~6 marks)

7. *(from 1.2.1)* Describe, in order, the steps the CPU and OS take to handle an **interrupt**, making clear why a **stack** is used. **[3]** *(AO1)*

8. *(from 1.1.2)* Explain **one** way in which a **RISC** processor differs from a **CISC** processor. **[1]** *(AO1)*

9. *(from 1.1.1)* State Von Neumann architecture's key principle that distinguishes it from Harvard architecture. **[2]** *(AO1)*